Project Proposal

SecureX Pro

Unified Vulnerability Assessment and Exploitation Analysis Framework

By

Saboor Saleem - 221395

Eman Fatima - 221372

Muhammad Talha Masood -221452

Supervisor

Muhammad Ayaz Khan

DEPARTMENT OF COMPUTER SCIENCE AIR UNIVERSITY ISLAMABAD (AU)

Table of Contents

RELATED SYSTEM ANALYSIS/ Literature Review:..... 3

PROBLEM STATEMENT: 4

PROBLEM SOLUTION/OBJECTIVES OF PROPOSED SYSTEM: 4

Objectives: 4

SCOPE: 5

MODULES: 5

System Limitations/Constraints: 6

DATA GATERING: 7

TOOLS AND TECHNOLOGY: 7

Project stakeholders and Roles: 8

Module-based Work Division: 8

GANTT CHART: 10

References :..... 10

SUPERVISOR MEETING LOG:

	DISCUSSION	TASK ASSIGNMENT
MEETING 1 DATE: 13-12-2025	Problem Identification& Feasibility Analysis	Research about the existing systems and how you will get the data
MEETING 2 DATE: 23-12-2025	Problem Analysis and Competitive Landscape	Document key challenges and initial system requirements.

INTRODUCTION:

With the increasing dependence on networked systems, cloud services, and web applications, cybersecurity threats have become more frequent, complex, and damaging. Organizations face continuous risks from vulnerabilities arising due to misconfigurations, outdated services, insecure web applications, and weak system defenses. Traditional security assessment processes rely on multiple independent tools such as Nmap, OpenVAS, Burp Suite, and Metasploit, which require expert knowledge to operate and interpret collectively.

Most existing tools perform isolated tasks such as reconnaissance, vulnerability scanning, or exploitation, but lack a unified intelligence layer that correlates findings, prioritizes risks, and provides actionable insights. Manual security assessments are time-consuming, prone to human error, and inefficient in large-scale environments.

To address these challenges, **SecureX Pro** is proposed as an **intelligent, unified vulnerability assessment and penetration testing framework**. The system automates reconnaissance, vulnerability analysis, exploit correlation, web application testing, risk assessment, and reporting within a single platform. By intelligently enhancing reconnaissance results and correlating them with vulnerability and exploit intelligence, the system improves detection accuracy, reduces false positives, and enables faster, data-driven security decision-making.

RELATED SYSTEM ANALYSIS/ Literature Review:

APPLICATION NAME	WEAKNESS	PROPOSED PROJECT SOLUTION
Nmap	Performs only reconnaissance, no vulnerability correlation	Enhances Nmap output with CVE and exploit intelligence mapping
Nessus	Expensive, limited exploitation context	Adds exploit feasibility analysis and risk prioritization
OpenVAS	Generates large uncorrelated reports	Intelligent vulnerability correlation and ranking
Metasploit	Exploitation- logic, no focused,requires manual analysis	Adds AI-based forecasting to improve arbitrage precision
Burp Suite	Web-focused only, isolated results	Unified web and network vulnerability analysis

Existing research emphasizes the need for automated vulnerability management systems that integrate multiple assessment phases. However, most tools still lack intelligent correlation and unified reporting, creating a gap that this project aims to fill.

PROBLEM STATEMENT:

Organizations and security analysts face difficulties in maintaining strong cybersecurity posture due to:

- Fragmented security tools
- Manual vulnerability correlation
- Time-consuming penetration testing
- High false-positive rates
- Lack of unified risk prioritization

- Absence of automated reporting
- Requirement of expert-level knowledge

These challenges necessitate an **unified vulnerability assessment solution** capable of automating the complete security testing lifecycle from reconnaissance to reporting.

PROBLEM SOLUTION/OBJECTIVES OF PROPOSED SYSTEM:

SecureX Pro addresses these issues by providing:

- Automated reconnaissance and scanning
- Intelligent vulnerability correlation
- Exploit feasibility analysis
- Integrated web application testing
- Risk scoring and prioritization
- Automated professional reporting
- Centralized dashboard for monitoring

This solution improves efficiency, accuracy, and accessibility of security assessments through intelligent automation.

Objectives:

General Objective:

To develop an intelligent, unified cybersecurity assessment platform that automates reconnaissance, vulnerability detection, exploit analysis, and reporting to enhance system security evaluation.

Specific Objectives:

- To automate network reconnaissance using Nmap
- To correlate discovered services with CVE databases
- To identify exploit availability and feasibility
- To integrate web vulnerability testing
- To implement risk scoring and prioritization
- To generate automated vulnerability reports
- To provide a centralized monitoring dashboard
- To ensure ethical testing within controlled environments

SCOPE:

The scope of SecureX Pro includes scanning local and test network environments, performing vulnerability assessments on virtual machines and intentionally vulnerable systems such as Metasploitable, DVWA, and OWASP Juice Shop. The system will analyze open ports, running services, web applications, and known vulnerabilities.

The project does **not** include real-world unauthorized penetration testing, zero-day exploit development, or attacks on live production systems.

MODULES:

1.Reconnaissance Module.

Purpose: Collects low-level network and system information from the target.

Functions:

Performs automated network scanning using Nmap.

Identifies:

- Live hosts
- Open TCP/UDP ports (e.g., 21, 22, 80, 443)
- Running services (Apache, SSH, FTP, MySQL, etc.)
- Service versions
- Operating system fingerprints (Linux, Windows – best effort)
- Produces structured scan output (XML/JSON)

2. Vulnerability Correlation Engine

Purpose: Converts reconnaissance data into known security weaknesses.

Functions:

Maps detected services and versions to known vulnerabilities Uses:

- NVD (National Vulnerability Database)
- CVE Details

Validates vulnerability applicability using:

- Exact version matching
- OS compatibility checks
- Eliminates false positives caused by generic CVEs

3.Exploit Intelligence & Analysis Module

Purpose: Evaluates exploit feasibility without performing real attacks.

Functions:

Identifies potential exploit types, such as:

- Service misconfiguration exploits
- Authentication bypass possibilities
- Information disclosure vulnerabilities
- Integrates Metasploit exploit metadata (search-only)

Analyses:

- Exploit availability
- Exploit complexity
- Required privileges
- Performs non-intrusive feasibility simulation

4. Web Vulnerability Assessment Module

Purpose: Identifies vulnerabilities in web applications and web services.

Functions:

Scans for OWASP Top 10 vulnerabilities:

1. Broken Access Control
2. Cryptographic Failures
3. Injection (SQL, Command)
4. Insecure Design
5. Security Misconfiguration
6. Vulnerable & Outdated Components
7. Authentication Failures
8. Software & Data Integrity Failures
9. Logging & Monitoring Failures
10. Server-Side Request Forgery (SSRF)

Integrates:

- OWASP ZAP (dynamic testing)
- Nikto (web server configuration checks)

Detects:

- SQL Injection
- XSS
- CSRF
- Missing security headers
- Insecure cookies

5.Risk Scoring & Prioritization Module

Purpose: Converts raw vulnerabilities into actionable risk insights.

Functions: Assigns severity using CVSS v3.1

Evaluates:

- Exploit availability
- Service exposure (internal vs external)
- Potential impact on CIA triad Categorizes vulnerabilities as:
- Low
- Medium
- High
- Critical

6 .Reporting & Remediation Module

Purpose: Presents findings in a professional, decision-ready format.

Functions:

- Generates automated
- PDF reports
- HTML reports

Includes:

- Executive summary
- Technical vulnerability details
- Risk severity charts
- Provides clear remediation recommendations, such as:
- Software updates
- Configuration hardening
- Service isolation

7. Logging, Auditing & Monitoring Module

Purpose: Ensures transparency, traceability, and system reliability.

Functions:

Logs:

- Scan initiation and completion
- Tool execution events
- User actions

Tracks errors:

- Tool execution failures
- API timeouts
- Database errors •

Detects anomalies:

- Abnormally long scans
- Repeated failed login attempts
- Maintains audit trails for accountability

8.User Authentication & Access Control Module

Purpose: Ensures only authorized users can access system features.

Functions:

Secure user registration and login

Password hashing and encryption

Role-based access: Admin

Regular User

Restricts sensitive operations based on role

9.Dashboard & Visualization Module

Purpose: Provides an interactive interface for users.

Functions:

Displays:

- Scan results
- Risk levels
- Vulnerability statistics

Features:

- Search by CVE, host, severity
- Filters by date, module, risk level
- Report download options

User-friendly and responsive design

10.**Scan Orchestration & Workflow Management Module.**

Purpose: Controls the overall scanning process and inter-module coordination.

Functions:

Manages scan lifecycle:

- Scan creation
- Execution
- Completion
- Coordinates data flow between modules **Ensures**:
- Correct execution order
- Resource management
- Scan isolation per user
- Handles concurrent scans safely

System Limitations/Constraints:

- Results depend on available vulnerability databases
- Limited to authorized test environments
- Cannot detect zero-day vulnerabilities
- Performance depends on network size
- API rate limitations of integrated tools

Data Gathering:

SecureX Pro gathers data from network scans, service detection results, vulnerability databases (CVE), exploit repositories, and web application scanners. Data is parsed, normalized, and stored in structured form to enable accurate correlation and analysis. All testing data is collected from legally authorized and controlled lab environments.

TOOLS AND TECHNOLOGY:

Category	Tools	Rationale
Programming	Python	Automation and tool integration
Reconnaissance	Nmap	Network discovery
Vulnerability Scan	OpenVAS	Vulnerability detection
Exploitation	Metasploit	Exploit intelligence
Web Testing	OWASP ZAP	Web vulnerability scanning
Backend	Flask / FastAPI	High-performance API
Database	SQLite / MongoDB	Data storage
Frontend	React js / next js	Dashboard UI
DevOps	Git+Github	Vision control

Project stakeholders and Roles:

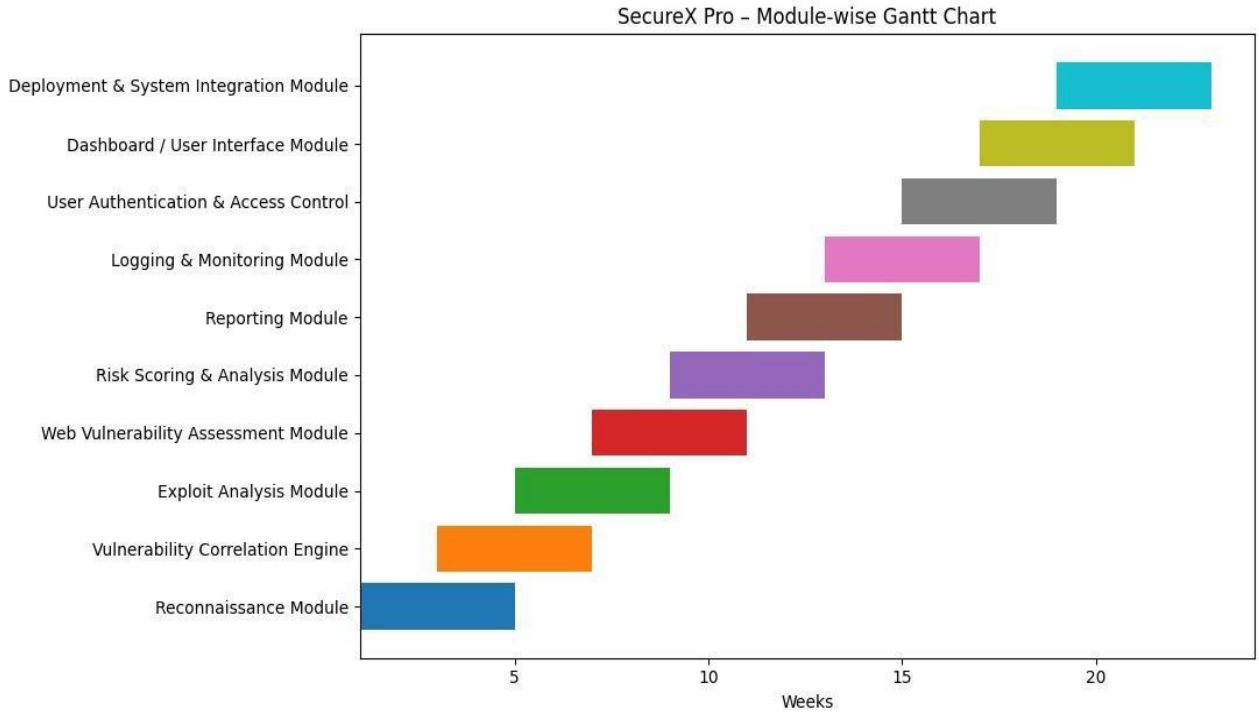
Project sponsor	Air University Islamabad.
Stakeholders	● Saboor Saleem ● Eman Fatima. ● Muhammad Talha Masood ● Muhammad Ayaz Khan ● Final Year Project Committee: Evaluation of project.

Module-based Work Division:

Student Name	**Registration Number**	**Module/responsibility**
M. Talha Masood	221452	**Reconnaissance Module** • Automated network scanning using Nmap.Live hosts, open ports, services, OS detection • Structured scan result storage **Vulnerability Correlation Engine** • CVE mapping for detected services. Exploit database matching • False positive reduction via version validation **Exploit Analysis Module** • Identification of potential exploits • Controlled exploit feasibility simulation • Integration with Metasploit intelligence

Eman Fatima	221374	**Web Vulnerability Assessment Module** • OWASP Top 10 vulnerability scanning • Integration of OWASP ZAP and Nikto • Detection of SQLi, XSS, CSRF, misconfigurations **Risk Scoring & Analysis Module** • CVSS-based severity scoring • Exploit availability and exposure analysis • Risk-based vulnerability prioritization **Reporting Module** • Automated PDF & HTML report generation • Remediation recommendations • Graphs, charts, and severity summaries
Saboor Saleem	221395	**Logging & Monitoring Module** • Scan and vulnerability logging • Error and anomaly tracking • Audit trail maintenance **User Authentication & Access Control** • Secure login and registration • Role-based access control • Encrypted credential storage **Dashboard / User Interface Module** • Visualization of scan results and risks • Real-time monitoring dashboard • Interactive and user-friendly UI **Scan Orchestration & Workflow Management Module.** Controls the overall scanning process and inter-module coordination.

GANTT CHART:



References:

- Scarfone, K., & Mell, P. (2012). Guide to Intrusion Detection and Prevention Systems. NIST
- Behl, A., & Behl, K. (2017). Cyberwar: The Next Threat to National Security. Oxford Press
- OWASP Foundation. OWASP Top 10, 2023
- Nmap Documentation, 2024
- Metasploit Framework Documentation, 2024
- CVE Database – MITRE